

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273605

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

MIYAHARA; Satoshi

SEMICONDUCTOR DEVICE

Abstract

An object is to provide a technology that can enhance the magnetic coupling effect. A semiconductor device includes: a P electrode including a first main surface portion and a first external connecting portion; an N electrode including a second main surface portion and a second external connecting portion; and a C electrode including a third main surface portion, and a third-first external connecting portion and a third-second external connecting portion. The first external connecting portion and the third-first external connecting portion are arranged in a direction in which the first main surface portion and the third main surface portion face each other, and the second external connecting portion and the third-second external connecting portion are arranged in a direction in which the second main surface portion and the third main surface portion face each other.

Inventors: MIYAHARA; Satoshi (Tokyo, JP)

Applicant: Mitsubishi Electric Corporation (Tokyo, JP)

Family ID: 1000008360476

Assignee: Mitsubishi Electric Corporation (Tokyo, JP)

Appl. No.: 18/979495

Filed: December 12, 2024

Foreign Application Priority Data

JP 2024-028595

Feb. 28, 2024

Publication Classification

Int. Cl.: H01L23/64 (20060101); H01L23/498 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present disclosure relates to a semiconductor device.

Description of the Background Art

[0002] What has been proposed is connecting, in a semiconductor device including a three-level circuit in one package, a three-level direct potential and a plurality of semiconductor elements with a plurality of electrodes and disposing main surfaces of the plurality of electrodes to face each other (e.g., Japanese Patent Application Laid-Open No. 2014-155287). Under this technology, a current passing through the main surfaces facing each other provides a magnetic coupling effect for reducing the inductance. This can suppress the surge voltage caused by interactions between abrupt current variations in switching the semiconductor elements and the inductance.

[0003] However, since the conventional art has not optimized positions of external connectors that are part of the electrodes and are connected to, for example, filter capacitors, there has been room for improvement on the magnetic coupling effect for reducing the inductance.

SUMMARY

[0004] The present disclosure has been made in view of the problem, and has an object of providing a technology that can enhance the magnetic coupling effect.

[0005] A semiconductor device includes: a first semiconductor element, a second semiconductor element, a third semiconductor element, and a fourth semiconductor element that are included in a three-level circuit; a P electrode including a first main surface portion and a first external connecting portion connected to a positive electrode of a DC circuit, the P electrode connecting the positive electrode to the first semiconductor element; an N electrode including a second main surface portion and a second external connecting portion connected to a negative electrode of the DC circuit, the N electrode connecting the negative electrode to the second semiconductor element; and a C electrode including a third main surface portion, and a third-first external connecting portion and a third-second external connecting portion that are connected to a neutral point of the DC circuit, the C electrode connecting the neutral point to the third semiconductor element, wherein the first main surface portion and the second main surface portion face and overlap the third main surface portion, the first external connecting portion and the third-first external connecting portion are arranged in a direction in which the first main surface portion and the third main surface portion face each other, and the second external connecting portion and the third-second external connecting portion are arranged in a direction in which the second main surface portion and the third main surface portion face each other.

[0006] This can enhance the magnetic coupling effect.

[0007] These and other objects, features, aspects, and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a circuit diagram illustrating a three-level circuit including a semiconductor device according to Embodiment 1;

[0009] FIG. 2 is a perspective view schematically illustrating a structure of the semiconductor device according to Embodiment 1;

[0010] FIG. 3A is a perspective view and FIGS. 3B and 3C are front elevation views all of which schematically illustrate a part of the structure of the semiconductor device according to Embodiment 1;

[0011] FIG. 4A is a perspective view and FIGS. 4B to 4D are front elevation views all of which schematically illustrate a part of a structure of a semiconductor device according to Embodiment 2;

[0012] FIG. 5A is a perspective view and FIGS. 5B to 5D are front elevation views all of which schematically illustrate a first relevant structure;

[0013] FIG. 6A is a perspective view and FIGS. 6B and 6C are front elevation views all of which schematically illustrate a second relevant structure;

[0014] FIG. 7A is a perspective view and FIGS. 7B and 7C are front elevation views all of which schematically illustrate a part of a structure of a semiconductor device according to Embodiment 3;

[0015] FIG. 8A is a perspective view and FIGS. 8B and 8C are front elevation views all of which schematically illustrate a part of a structure of a semiconductor device according to Embodiment 4; and

[0016] FIG. 9A is a perspective view and FIGS. 9B to 9D are front elevation views all of which schematically illustrate a part of a structure of a semiconductor device according to Embodiment 5.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0017] Embodiments will be described with reference to the attached drawings. The features to be described in Embodiments below are mere exemplifications, and all of the features are not necessarily essential. In the description below, identical constituent elements in a plurality of Embodiments will be denoted by the same or similar reference numerals, and different constituent elements will be mainly described. In the following description, a particular position and a particular direction such as “up”, “down”, “left”, and “right” need not always coincide with an actual position and an actual direction.

Embodiment 1

[0018] FIG. 1 is a circuit diagram illustrating a three-level circuit including a semiconductor device according to Embodiment 1. The semiconductor device according to Embodiment 1 includes a first semiconductor element **111**, a second semiconductor element **112**, a third semiconductor element **113**, a fourth semiconductor element **114**, a P electrode P, an N electrode N, a C electrode C, an AC electrode AC, and an insulating substrate and a package to be described later.

[0019] The first semiconductor element **111**, the second semiconductor element **112**, the third semiconductor element **113**, and the fourth semiconductor element **114** include insulated-gate bipolar transistors (IGBTs) **111a**, **112a**, **113a**, and **114a**, and freewheeling diodes (FWDs) **111b**, **1112b**, **113b**, and **114b** anti-parallel connected to the IGBTs **111a**, **112a**, **113a**, and **114a**, respectively.

[0020] The first semiconductor element **111** to the fourth semiconductor element **114** may include metal-oxide-semiconductor field-effect transistors (MOSFETs) or reverse conducting IGBTs (RC-IGBTs), instead of the IGBTs **111a**, **112a**, **113a**, and **114a**. The FWDs **111b** to **114b** are, for example, Schottky barrier diodes (SBD) or p-n junction diodes (PND).

[0021] The first semiconductor element **111** to the fourth semiconductor element **114** may be made of silicon (Si), or a wide-bandgap semiconductor such as silicon carbide (SiC), gallium nitride (GaN), gallium oxide (Ga.sub.2O.sub.3), or diamond. The first semiconductor element **111** to the fourth semiconductor element **114** which are made of the wide-bandgap semiconductor can perform stable operations at high temperatures and high voltages, and accelerate switching speeds. Each of the first semiconductor element **111** to the fourth semiconductor element **114** may be a single semiconductor chip, or include a plurality of semiconductor chips connected in parallel.

[0022] An end of a pair of filter capacitors **301** connected in series corresponds to a positive electrode of a DC circuit, and has a positive potential **121**. The other end of the pair of filter

capacitors **301** connected in series corresponds to a negative electrode of the DC circuit, and has a negative potential **122**. A connecting portion of the filter capacitors **301** corresponds to a neutral point of the DC circuit, and has a neutral-point potential **123**. A connecting point **115** that connects the first semiconductor element **111**, the second semiconductor element **112**, and the fourth semiconductor element **114** has a AC potential **124**.

[0023] A collector of the IGBT **111a** is connected to the positive electrode having the positive potential **121**. An emitter of the IGBT **111a** is connected to a collector of the IGBT **112a** through the connecting point **115**. An emitter of the IGBT **112a** is connected to the negative electrode having the negative potential **122**. As described above, the first semiconductor element **111** and the second semiconductor element **112** are connected in series between the positive electrode having the positive potential **121** and the negative electrode having the negative potential **122**.

[0024] An emitter of the IGBT **113a** is connected to the neutral point having the neutral-point potential **123**. A collector of the IGBT **113a** is connected to a collector of the IGBT **114a**. An emitter of the IGBT **114a** is connected to the connecting point **115** of the first semiconductor element **111** and the second semiconductor element **112**. As described above, the third semiconductor element **113** and the fourth semiconductor element **114** are connected in series between the neutral point having the neutral-point potential **123** and the connecting point **115**.

[0025] The third semiconductor element **113** and the fourth semiconductor element **114** appropriately select voltages clamped by the filter capacitors **301**, so that the three-level circuit with the aforementioned structure outputs stepping voltages. Although the semiconductor device is described above as a T-shaped three-level circuit in which the first semiconductor element **111** to the fourth semiconductor element **114** are connected in a T shape, the semiconductor device is not limited to this. For example, the semiconductor device may be an I-shaped three-level circuit in which four semiconductor elements (e.g., the first semiconductor element **111** to the fourth semiconductor element **114**) are connected in an I shape and another two semiconductor elements (e.g., fifth and sixth semiconductor elements) are connected to a neutral point. In this case, the first semiconductor element **111** to the fourth semiconductor element **114** are connected in series between a positive electrode and a negative electrode. Then, the fifth semiconductor element is connected between the neutral point and a connecting point of the first semiconductor element **111** and the second semiconductor element **112**, and the sixth semiconductor element is connected between the neutral point and a connecting point of the third semiconductor element **113** and the fourth semiconductor element **114**.

[0026] FIG. **2** is a perspective view schematically illustrating a structure of the semiconductor device according to Embodiment 1. Conductive patterns **272** are disposed on an insulating substrate **271**, and the first semiconductor element **111** to the fourth semiconductor element **114**, which are not illustrated in FIG. **2**, are disposed on the conductive patterns **272**. The conductive patterns **272** are selectively connected by wires that are not illustrated to complete the circuit in FIG. **1**.

[0027] Not only the P electrode P, the N electrode N, and the C electrode C but also the AC electrode AC that connects the AC potential **124** not illustrated in FIG. **2** to an external output circuit are disposed on the conductive patterns **272**. The structure in FIG. **2** is almost covered with a package that is not illustrated.

[0028] FIG. **3A** is a perspective view and FIGS. **3B** and **3C** are front elevation views all of which schematically illustrate a structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 1.

[0029] As illustrated in FIGS. **3A** and **3B**, the P electrode P includes a P main surface portion **221** that is a first main surface portion, a P external connecting portion **231** that is a first external connecting portion, and a P internal connecting portion **251** that is a first internal connecting portion.

[0030] The P external connecting portion **231** is formed at an upper portion of the P main surface

portion **221**, and is connected to the positive electrode having the positive potential **121** in FIG. **1** through an external wire such as a bus bar. The P external connecting portion **231** is provided with an attachment hole **241** for connecting the external wire via a screw. The P internal connecting portion **251** is formed at a lower portion of the P main surface portion **221**, and is connected to the first semiconductor element **111** in FIG. **1** through the conductive pattern **272** in FIG. **2**. Although the number of each of the P external connecting portions **231** and the P internal connecting portions **251** is one in Embodiment 1, the number may be multiple.

[0031] As illustrated in FIGS. **3A** and **3B**, the N electrode N includes an N main surface portion **222** that is a second main surface portion, an N external connecting portion **232** that is a second external connecting portion, and an N internal connecting portion **252** that is a second internal connecting portion.

[0032] The N external connecting portion **232** is formed at an upper portion of the N main surface portion **222**, and is connected to the negative electrode having the negative potential **122** in FIG. **1** through an external wire such as a bus bar. The N external connecting portion **232** is provided with an attachment hole **242** for connecting the external wire via a screw. The N internal connecting portion **252** is formed at a lower portion of the N main surface portion **222**, and is connected to the second semiconductor element **112** in FIG. **1** through the conductive pattern **272** in FIG. **2**.

Although the number of each of the N external connecting portions **232** and the N internal connecting portions **252** is one in Embodiment 1, the number may be multiple.

[0033] As illustrated in FIGS. **3A** and **3C**, the C electrode C includes a C main surface portion **223** that is a third main surface portion, a first C external connecting portion **233a** that is a third-first external connecting portion, a second C external connecting portion **233b** that is a third-second external connecting portion, and a C internal connecting portion **253** that is a third internal connecting portion.

[0034] The first C external connecting portion **233a** and the second C external connecting portion **233b** are formed at upper portions of the C main surface portion **223**, and are connected to the neutral point having the neutral-point potential **123** in FIG. **1** through external wires such as bus bars. The first C external connecting portion **233a** and the second C external connecting portion **233b** are provided with an attachment hole **243a** and an attachment hole **243b**, respectively, for connecting the external wires via screws. The C internal connecting portion **253** is formed at a lower portion of the C main surface portion **223**, and is connected to the third semiconductor element **113** in FIG. **1** through the conductive pattern **272** in FIG. **2**.

[0035] Although the number of each of the first C external connecting portions **233a** and the second C external connecting portions **233b** is one in Embodiment 1, the number is not limited to this. The number of the first C external connecting portions **233a** should be identical to that of the P external connecting portions **231**, and the number of the second C external connecting portions **233b** should be identical to that of the N external connecting portions **232**. Although the number of the C internal connecting portions **253** is one in Embodiment 1, the number may be multiple as will be described later in Embodiment 3.

[0036] As illustrated in FIG. **3A**, the P main surface portion **221** and the N main surface portion **222** face and overlap the C main surface portion **223**. In Embodiment 1, the P main surface portion **221** and the N main surface portion **222** face one surface of the C main surface portion **223**, and the P main surface portion **221** and the N main surface portion **222** are in close proximity to each other. Each of the P main surface portion **221** to the C main surface portion **223** may be almost planar, and partly bent to facilitate connection to the conductive patterns **272** and the bus bars.

[0037] The P external connecting portions **231** and the first C external connecting portions **233a** are arranged in a direction **262** in which the P main surface portion **221** and the C main surface portion **223** face each other. The positions of the P external connecting portions **231** and the first C external connecting portions **233a** are identical in a horizontal axis **261**. The positions of the attachment hole **241** and the attachment hole **243a** are identical in the horizontal axis **261**.

[0038] The N external connecting portion **232** and the second C external connecting portion **233b** are arranged in a direction **263** in which the N main surface portion **222** and the C main surface portion **223** face each other. The positions of the N external connecting portion **232** and the second C external connecting portion **233b** are identical in the horizontal axis **261**. The positions of the attachment hole **242** and the attachment hole **243b** are identical in the horizontal axis **261**.

[0039] In Embodiment 1, the P external connecting portion **231** and the N external connecting portion **232** are aligned in a row along the horizontal axis **261**, and the first C external connecting portion **233a** and the second C external connecting portion **233b** are aligned in another row along the horizontal axis **261**.

[0040] When the semiconductor device includes multiple pairs of the P external connecting portion **231** and the first C external connecting portion **233a**, the P external connecting portion **231** and the first C external connecting portion **233a** in each of the pairs are preferably arranged in the direction **262**. Similarly, when the semiconductor device includes multiple pairs of the N external connecting portion **232** and the second C external connecting portion **233b**, the N external connecting portion **232** and the second C external connecting portion **233b** in each of the pairs are preferably arranged in the direction **263**. The P external connecting portion **231** and the N external connecting portion **232** may be or need not be in close proximity to each other.

Summary of Embodiment 1

[0041] In Embodiment 1, the P main surface portion **221** and the N main surface portion **222** face and overlap the C main surface portion **223**. Since the magnetic coupling effect in such a structure can reduce the inductance each between the positive electrode and the neutral point and between the neutral point and the negative electrode, the surge voltage occurring in switching the semiconductor elements can be reduced.

[0042] In Embodiment 1, the P external connecting portions **231** and the first C external connecting portions **233a** are arranged in the direction **262**. The N external connecting portion **232** and the second C external connecting portion **233b** are arranged in the direction **263**. Since inlets and outlets of current paths of the P electrode and the C electrode C overlap each other and inlets and outlets of current paths of the C electrode C and the N electrode N overlap each other in such a structure, the magnetic coupling effect can be enhanced, and the inductance and the surge voltage can be further reduced. Similarly, since overlapping of inlets and outlets of current paths of the bus bars can be expected, additional enhancement of the magnetic coupling effect can also be expected.

[0043] In Embodiment 1, the P external connecting portion **231** and the N external connecting portion **232** are aligned in a row, and the first C external connecting portion **233a** and the second C external connecting portion **233b** are aligned in another row. Such a structure can shape the bus bar to be connected to the first C external connecting portion **233a** and the second C external connecting portion **233b** into a simple plate. This can consequently simplify manufacturing of external wires and reduce the cost.

Embodiment 2

[0044] FIG. **4A** is a perspective view and FIGS. **4B** to **4D** are front elevation views all of which schematically illustrate a structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 2.

[0045] In Embodiment 2, the P external connecting portion **231** and the second C external connecting portion **233b** are aligned in a row along the horizontal axis **261**, and the N external connecting portion **232** and the first C external connecting portion **233a** are aligned in another row along the horizontal axis **261**.

[0046] Consequently, the P main surface portion **221** can face the first surface of the C main surface portion **223**, and the N main surface portion **222** can face the second surface on the other side of the first surface of the C main surface portion **223**. In other words, a region of the P main surface portion **221** which overlaps the C main surface portion **223** and a region of the N main surface portion **222** which overlaps the C main surface portion **223** can be increased. This can

enhance the magnetic coupling effect, and further reduce the inductance in each of the current path of the P electrode P and the C electrode C and the current path of the C electrode C and the N electrode N. Thus, the surge voltage can be further reduced.

[0047] As illustrated in the front elevation views of the P electrode P, the N electrode N, and the C electrode C, the C internal connecting portion **253** is closer to the center of each of the main surface portions than the P internal connecting portion **251** and the N internal connecting portion **252** in Embodiment 2.

[0048] FIG. 5A is a perspective view and FIGS. 5B to 5D are front elevation views all of which schematically illustrate a structure (hereinafter referred to as a “first relevant structure”) which is relevant to the structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 2. In the first relevant structure, the P internal connecting portion **251** or the N internal connecting portion **252** is closer to the center of each of the main surface portions than the C internal connecting portion **253** as illustrated in the front elevation view of the P electrode P, the N electrode N, and the C electrode C of FIG. 5D.

[0049] In the structure of FIGS. 5A to 5D, a distance **611** between the P internal connecting portion **251** and the C internal connecting portion **253** becomes smaller than a distance **612** between the C internal connecting portion **253** and the N internal connecting portion **252**. Thus, a difference between the magnetic coupling effect between the P electrode P and the C electrode C and the magnetic coupling effect between the C electrode C and the N electrode N becomes large. This results in an increase in a difference between the inductance of the current path of the P electrode P and the C electrode C and the inductance of the current path of the C electrode C and the N electrode N. Thus, a difference between the surge voltages occurring in different operation modes is increased.

[0050] In contrast, the C internal connecting portion **253** is closer to the center of each of the main surface portions than the P internal connecting portion **251** and the N internal connecting portion **252** in a front view in Embodiment 2, as illustrated in the front elevation view of the P electrode P, the N electrode N, and the C electrode C in FIG. 4D. Such a structure can reduce a difference between a distance **613** between the P internal connecting portion **251** and the C internal connecting portion **253** and a distance **614** between the C internal connecting portion **253** and the N internal connecting portion **252**. Since this can reduce the difference between the surge voltages occurring in the different operation modes, optimization of driving conditions for each of the operation modes becomes unnecessary or is facilitated.

Embodiment 3

[0051] FIG. 6A is a perspective view and FIGS. 6B and 6C are front elevation views all of which schematically illustrate a structure (hereinafter referred to as a “second relevant structure”) which is relevant to the structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 3. In the second relevant structure, one of a plurality of P internal connecting portions **251** or a plurality of N internal connecting portions **252** is closer to the center of each of the main surface portions than a plurality of C internal connecting portions **253** in a front view.

[0052] In such a structure, current paths of currents which flow from the C internal connecting portions **253** pass outside a current region **711** of a current flowing through the P electrode P as indicated by arrows in FIGS. 6B and 6C. Since a region in which the current region **711** flowing through the P electrode P overlaps the current region **711** flowing through the C electrode C is small, the effect of reducing the inductance by the magnetic coupling becomes smaller.

[0053] FIG. 7A is a perspective view and FIGS. 7B and 7C are front elevation views all of which schematically illustrate a structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 3. In Embodiment 3, the plurality of C internal connecting portions **253** are closer to the center of each of the main surface portions than the plurality of P internal connecting portions **251** and the plurality of N internal connecting portions **252** in a front view. Since a region in which a current region **811** flowing through the P electrode P overlaps the current

region **811** flowing through the C electrode C is large in such a structure, the effect of reducing the inductance by the magnetic coupling between the P electrode P and the C electrode C can be enhanced. Thus, the surge voltage can be reduced. This holds true for the magnetic coupling between the C electrode C and the N electrode N.

Embodiment 4

[0054] FIG. **8A** is a perspective view and FIGS. **8B** and **8C** are front elevation views all of which schematically illustrate a structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 4. In Embodiment 4, a slit **911** extends in a horizontal direction and is provided in the P main surface portion **221**. One or more of the slits **911** should extend in the horizontal direction and be provided in at least one of the P main surface portion **221**, the N main surface portion **222**, or the C main surface portion **223**. In this Description, for example, at least one of A, B, C, . . . , or Z means any one of all combinations obtained by combining one type or more extracted from each of groups of A, B, C, . . . , and Z.

[0055] In such a structure, the slit **911** changes current paths of currents such that the current paths pass through the vicinity of the center of the electrode with the slit **911** as indicated by arrows in FIGS. **8B** and **8C**. This can, for example, increase a region in which a current region of a current flowing through the P electrode P overlaps a current region of a current flowing through the C electrode C, or increase an amount of a current passing through the overlapping region. Since the effect of reducing the inductance by the magnetic coupling between the P electrode P and the C electrode C can be enhanced, the surge voltage can be reduced. This holds true for the magnetic coupling between the C electrode C and the N electrode N.

Embodiment 5

[0056] FIG. **9A** is a perspective view and FIGS. **9B** to **9D** are front elevation views all of which schematically illustrate a structure of the P electrode P, the N electrode N, and the C electrode C according to Embodiment 5. In Embodiment 5, one or more of the P external connecting portions **231** (one in FIGS. **9A** and **9B**) are, for example, line-symmetric about a vertical axis **1011** on a plane of the electrode as a target axis as a whole. In other words, one or more of the P external connecting portions **231** (one in FIGS. **9A** and **9B**) are line-symmetric or substantially line-symmetric about the vertical axis **1011** as a target axis as a whole.

[0057] Similarly, one or more of the N external connecting portions **232** (two in FIGS. **9A** and **9C**), one or more of the first C external connecting portions **233a** (one in FIGS. **9A** and **9D**), and one or more of the second C external connecting portions **233b** (two in FIGS. **9A** and **9D**) are line-symmetric as a whole.

[0058] Such a structure can reduce deviations of current paths of currents flowing through the P electrode P and the N electrode N. Since the effect of reducing the inductance by the magnetic coupling with the C electrode C can be enhanced, the surge voltage can be reduced. Since the difference in inductance between the current paths can be reduced, the current balance between a plurality of chips can be enhanced, and a deviation in chip temperature caused by the current balance can be reduced.

[0059] Embodiments and the modifications can be freely combined, and appropriately modified or omitted.

[0060] A summary of various aspects of the present disclosure will be hereinafter described as Appendixes.

Appendix 1

A semiconductor device, comprising: [0061] a first semiconductor element, a second semiconductor element, a third semiconductor element, and a fourth semiconductor element that are included in a three-level circuit; [0062] a P electrode including a first main surface portion and a first external connecting portion connected to a positive electrode of a DC circuit, the P electrode connecting the positive electrode to the first semiconductor element; [0063] an N electrode including a second main surface portion and a second external connecting portion connected to a

negative electrode of the DC circuit, the N electrode connecting the negative electrode to the second semiconductor element; and [0064] a C electrode including a third main surface portion, and a third-first external connecting portion and a third-second external connecting portion that are connected to a neutral point of the DC circuit, the C electrode connecting the neutral point to the third semiconductor element, [0065] wherein the first main surface portion and the second main surface portion face and overlap the third main surface portion, [0066] the first external connecting portion and the third-first external connecting portion are arranged in a direction in which the first main surface portion and the third main surface portion face each other, and [0067] the second external connecting portion and the third-second external connecting portion are arranged in a direction in which the second main surface portion and the third main surface portion face each other.

Appendix 2

The semiconductor device according to appendix 1, [0068] wherein the first semiconductor element and the second semiconductor element are connected in series between the positive electrode and the negative electrode, and [0069] the third semiconductor element and the fourth semiconductor element are connected in series between the neutral point and a connecting point of the first semiconductor element and the second semiconductor element.

Appendix 3

The semiconductor device according to appendix 1, further comprising [0070] a fifth semiconductor element and a sixth semiconductor element that are included in the three-level circuit, [0071] wherein the first semiconductor element, the second semiconductor element, the third semiconductor element, and the fourth semiconductor element are connected in series between the positive electrode and the negative electrode, [0072] the fifth semiconductor element is connected between the neutral point and a connecting point of the first semiconductor element and the second semiconductor element, and [0073] the sixth semiconductor element is connected between the neutral point and a connecting point of the third semiconductor element and the fourth semiconductor element.

Appendix 4

The semiconductor device according to any one of appendixes 1 to 3, [0074] wherein the first external connecting portion and the second external connecting portion are aligned in a row, and [0075] the third-first external connecting portion and the third-second external connecting portion are aligned in an other row.

Appendix 5

The semiconductor device according to any one of appendixes 1 to 3, [0076] wherein the first external connecting portion and the third-second external connecting portion are aligned in a row, and [0077] the second external connecting portion and the third-first external connecting portion are aligned in an other row.

Appendix 6

The semiconductor device according to any one of appendixes 1 to 5, [0078] wherein the P electrode, the N electrode, and the C electrode further include a first internal connecting portion, a second internal connecting portion, and a third internal connecting portion to be connected to the first semiconductor element, the second semiconductor element, and the third semiconductor element, respectively, and [0079] the third internal connecting portion is closer to a center than the first internal connecting portion and the second internal connecting portion in a front view.

Appendix 7

The semiconductor device according to appendix 6, [0080] wherein a plurality of the third internal connecting portions are closer to a center than a plurality of the first internal connecting portions and a plurality of the second internal connecting portions in a front view.

Appendix 8

The semiconductor device according to any one of appendixes 1 to 7, [0081] wherein a slit extends

in a horizontal direction and is provided in at least one of the first main surface portion, the second main surface portion, or the third main surface portion.

Appendix 9

The semiconductor device according to any one of appendixes 1 to 8, [0082] wherein one or more of the first external connecting portions are line-symmetric as a whole, one or more of the second external connecting portions are line-symmetric as a whole, and one or more of the third-first external connecting portions and one or more of the third-second external connecting portions are line-symmetric as a whole.

[0083] While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

Claims

1. A semiconductor device, comprising: a first semiconductor element, a second semiconductor element, a third semiconductor element, and a fourth semiconductor element that are included in a three-level circuit; a P electrode including a first main surface portion and a first external connecting portion connected to a positive electrode of a DC circuit, the P electrode connecting the positive electrode to the first semiconductor element; an N electrode including a second main surface portion and a second external connecting portion connected to a negative electrode of the DC circuit, the N electrode connecting the negative electrode to the second semiconductor element; and a C electrode including a third main surface portion, and a third-first external connecting portion and a third-second external connecting portion that are connected to a neutral point of the DC circuit, the C electrode connecting the neutral point to the third semiconductor element, wherein the first main surface portion and the second main surface portion face and overlap the third main surface portion, the first external connecting portion and the third-first external connecting portion are arranged in a direction in which the first main surface portion and the third main surface portion face each other, and the second external connecting portion and the third-second external connecting portion are arranged in a direction in which the second main surface portion and the third main surface portion face each other.
2. The semiconductor device according to claim 1, wherein the first semiconductor element and the second semiconductor element are connected in series between the positive electrode and the negative electrode, and the third semiconductor element and the fourth semiconductor element are connected in series between the neutral point and a connecting point of the first semiconductor element and the second semiconductor element.
3. The semiconductor device according to claim 1, further comprising a fifth semiconductor element and a sixth semiconductor element that are included in the three-level circuit, wherein the first semiconductor element, the second semiconductor element, the third semiconductor element, and the fourth semiconductor element are connected in series between the positive electrode and the negative electrode, the fifth semiconductor element is connected between the neutral point and a connecting point of the first semiconductor element and the second semiconductor element, and the sixth semiconductor element is connected between the neutral point and a connecting point of the third semiconductor element and the fourth semiconductor element.
4. The semiconductor device according to claim 1, wherein the first external connecting portion and the second external connecting portion are aligned in a row, and the third-first external connecting portion and the third-second external connecting portion are aligned in an other row.
5. The semiconductor device according to claim 1, wherein the first external connecting portion and the third-second external connecting portion are aligned in a row, and the second external connecting portion and the third-first external connecting portion are aligned in an other row.
6. The semiconductor device according to claim 1, wherein the P electrode, the N electrode, and the

C electrode further include a first internal connecting portion, a second internal connecting portion, and a third internal connecting portion to be connected to the first semiconductor element, the second semiconductor element, and the third semiconductor element, respectively, and the third internal connecting portion is closer to a center than the first internal connecting portion and the second internal connecting portion in a front view.

7. The semiconductor device according to claim 6, wherein a plurality of the third internal connecting portions are closer to a center than a plurality of the first internal connecting portions and a plurality of the second internal connecting portions in a front view.

8. The semiconductor device according to claim 1, wherein a slit extends in a horizontal direction and is provided in at least one of the first main surface portion, the second main surface portion, or the third main surface portion.

9. The semiconductor device according to claim 1, wherein one or more of the first external connecting portions are line-symmetric as a whole, one or more of the second external connecting portions are line-symmetric as a whole, and one or more of the third-first external connecting portions and one or more of the third-second external connecting portions are line-symmetric as a whole.
